Project 3 - DSA 8020

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Contents

Project Description

The ERA-Interim is a global atmospheric reanalysis dataset. Reanalysis refers to the process of producing spatially and temporally gridded datasets through data assimilation, primarily for climate monitoring and analysis. In the R data file **NA__MaxT__ERA.RData**, you will find daily maximum temperature values from *1979* to *2017*, with a spatial resolution of approximately 80 km. The original global dataset has been subsetted to a region covering the contiguous United States and extending into Canada and Mexico.

In Problems 1 and 2, you are required to analyze an ERA-Interim average temperature time series at yearly and monthly temporal resolutions at a spatial location of your choice.

Problem 1. Analyzing Yearly Average Temperatures Time-Series

1.1. The following steps are needed to extract the annual mean temperatures at the specified location from the original dataset with daily temperatures at thousands of locations:

a. First, use the following R script to load the R data object:

```r
load("NA_MaxT_ERA.RData")
NA_MaxT <- NA_MaxT - 273.15 # change from Kelvin to Celsius
nlon <- length(NA_lon); nlat <- length(NA_lat)
nmon <- 12; nyr <- 39 # numbers of month/year
```

b. Second, choose a location of interest (with the information of longitude and latitude) and identify the nearest ERA grid cell to the chosen location. Below you can find a script to get the correct ERA grid cell for Clemson:

```r
# Clemson Example
Clemson.lon.lat <- c(-82.8374, 34.6834) # (lon,lat)
dist2Clemson <- rdist.earth(
```

```r
  matrix(Clemson.lon.lat, 1, 2),
  expand.grid(NA_lon, NA_lat),
  miles = F
  )
id <- which.min(dist2Clemson)
(lon_id <- id %% nlon)
```

```
## [1] 58
```

```r
(lat_id <- id %/% nlon + 1)
```

```
## [1] 15
```

```r
# Check
(rdist.earth(matrix(Clemson.lon.lat, 1, 2),
matrix(c(NA_lon[lon_id], NA_lat[lat_id]), 1, 2), miles = F)
== min(dist2Clemson))
```

```
##      [,1]
## [1,] TRUE
```

```r
Monterey.lon.lat <- c(-121.8979, 36.5973) # (lon,lat)
dist2Monterey <- rdist.earth(
  matrix(Monterey.lon.lat, 1, 2),
  expand.grid(NA_lon, NA_lat),
  miles = F
  )

id <- which.min(dist2Monterey)

(lon_id <- id %% nlon)
```

```
## [1] 5
```

```r
(lat_id <- id %/% nlon + 1)
```

```
## [1] 18
```

```r
(rdist.earth(
  matrix(Monterey.lon.lat, 1, 2),
  matrix( c(NA_lon[lon_id], NA_lat[lat_id]), 1, 2),
  miles = F
  )== min(dist2Monterey))
```

```
##      [,1]
## [1,] TRUE
```

c. Third, aggregate the daily data to the annual average time series. You could use the following script to achieve this:

```r
time_series_daily <- NA_MaxT[lon_id, lat_id,]
print("Daily Time Series summary: ")
```

```
## [1] "Daily Time Series summary: "
```

```r
print(summary(time_series_daily))
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   4.173  13.178  14.816  14.864  16.539  24.845
```

```r
time_series_yearly <- tapply(time_series_daily, list(yr), mean)
print("Yearly Time Series summary: ")
```

```
## [1] "Yearly Time Series summary: "
```

```r
print(summary(time_series_yearly))
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   14.06   14.44   14.74   14.86   15.26   16.18
```

1.2. Describe the main features of the annual time series data you extracted.

```r
# tsfeatures for (feature autoextraction)
yearly_features <- tsfeatures(time_series_yearly)
print(yearly_features)
```

```
## # A tibble: 1 x 16
##   frequency nperiods seasonal_period trend    spike linearity curvature e_acf1
##       <dbl>    <dbl>           <dbl> <dbl>    <dbl>     <dbl>     <dbl>  <dbl>
## 1         1        0               1 0.434 0.000433    0.0857      2.28 0.0822
## # i 8 more variables: e_acf10 <dbl>, entropy <dbl>, x_acf1 <dbl>,
## #   x_acf10 <dbl>, diff1_acf1 <dbl>, diff1_acf10 <dbl>, diff2_acf1 <dbl>,
## #   diff2_acf10 <dbl>
```

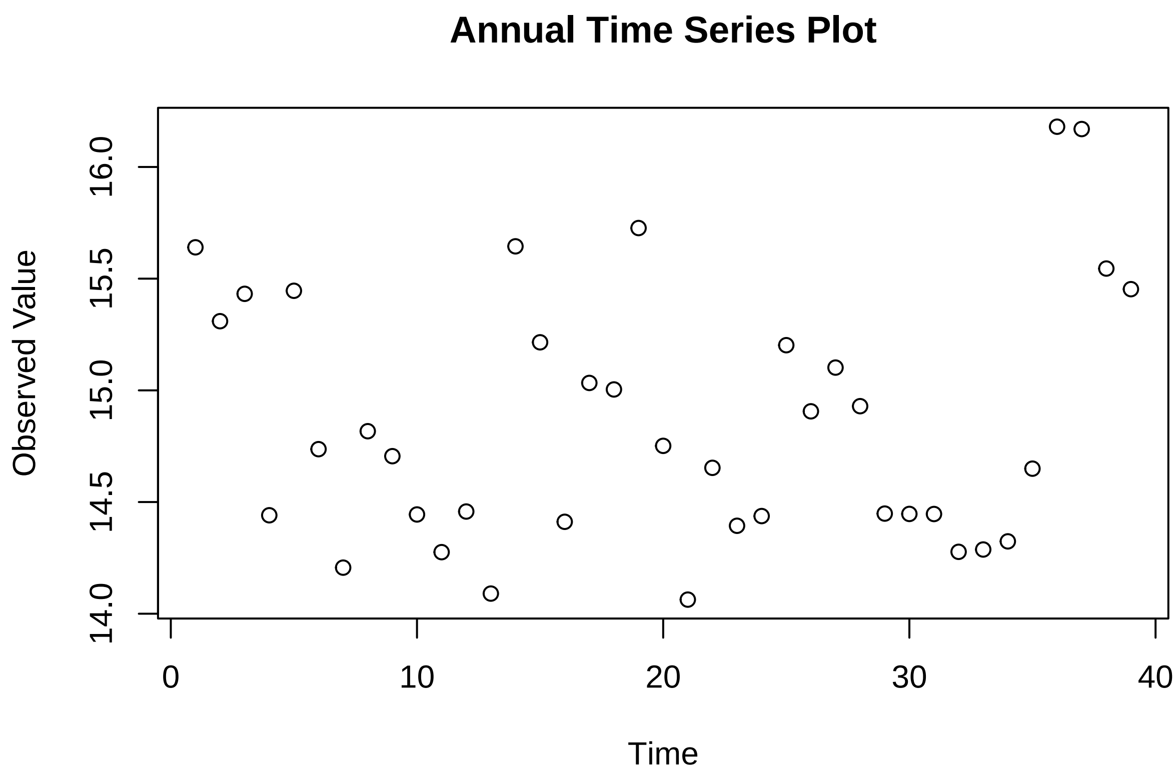
Our time series feature analysis using *tsfeatures* for the annual series reveals the following:

Feature	Value	Analysis
trend	0.4337	Moderately strong trend, indicates presence of long term pattern
spike	0.00042	Extremely low, indicates no significant anomalies or jumps in any given year
linearity	0.08572	Moderately low - some possibility model is linear
curvature	2.27771	High value strongly indicates non-linearity

Feature	Value	Analysis
entropy	0.9574	High value, indicates limited predictability and complex relations between the data
x_acf1	0.4117	Moderate indication of autocorrelation, any given years values show to depend on its previous year

1.3. Is it reasonable to assume that there is a linear trend? If so, estimate and remove the trend to get the detrended time series.

```
plot(time_series_yearly, main= "Annual Time Series Plot", ylab= "Observed Value", xlab= "Time")
```



Our graph does not reveal much significant information. However, based on our previous results from *tsfeatures*, it is still somewhat reasonable to expect the possibility of a linear trend:

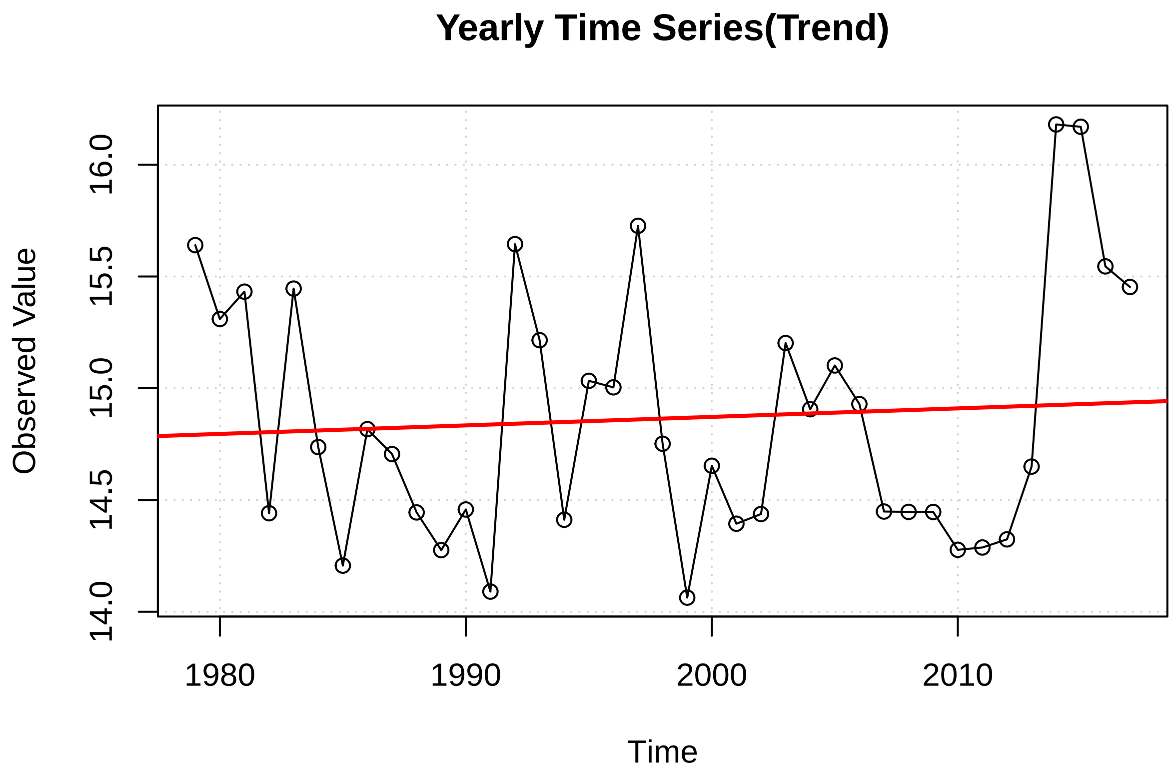
```
1. trend: value 0.43 shows significant enough trend in data to expect some prediction.

2. linearity: while this is not 0, it is still low enough to keep linearity as a reasonable possibility

3. curvature: even though value is high (indicating a non-perfect linear fit), it is still significant
```

Since a linear trend is not out of question for our time series, we will now estimate its value.

```r
yrs <- unique(yr)

# fitting linear model with trend
lm_annual <- lm(time_series_yearly ~ yrs)
summary(lm_annual)
```

```
##
## Call:
## lm(formula = time_series_yearly ~ yrs)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -0.8047 -0.4528 -0.1126  0.4420  1.2551
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  7.254522  16.272999   0.446    0.658
## yrs          0.003809   0.008145   0.468    0.643
##
## Residual standard error: 0.5724 on 37 degrees of freedom
## Multiple R-squared:  0.005876,   Adjusted R-squared:  -0.02099
## F-statistic: 0.2187 on 1 and 37 DF,  p-value: 0.6428
```

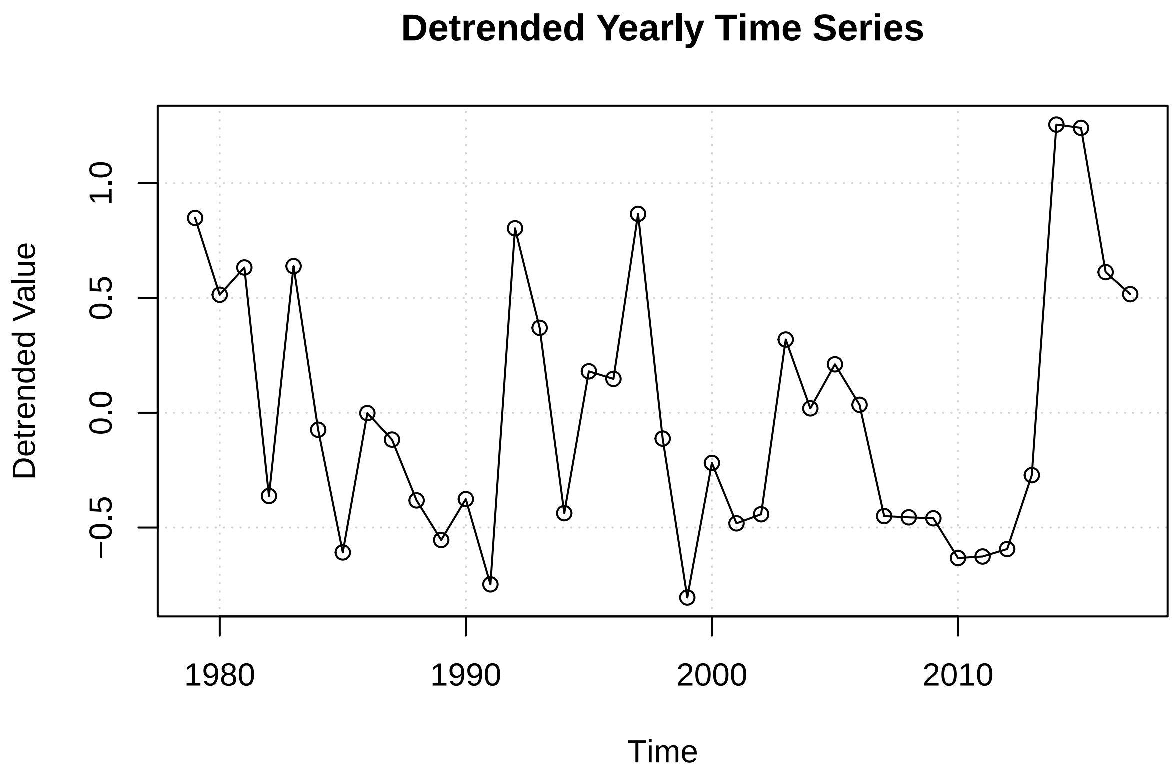
```r
plot(
  x= yrs,
  y= time_series_yearly,
  type= "o", # measured points w/ time continuity
  main= "Yearly Time Series(Trend)",
  ylab= "Observed Value",
  xlab= "Time",
  panel.first= grid()
  )
abline(lm_annual, col= "red", lwd= 2)
```



Our linear model fit reveals a p-value of 0.64, showing that a linear trend is statistically insignificant - leading us to fail to reject the null hypothesis and conclude there is no strong proof of linear behavior in the data. This is further confirmed by our Yearly Time Series(Trend) plot, which shows the data fluctuating significantly around our linear trend line (red) as well as many clearly non-linear patterns. It may be possible that the data could have a different measurable trend such as cyclical behavior, as is not unusual with environmental data.

Although our linear trend model was not statistically significant, we will proceed to detrend the time series data to obtain residual series which reflect deviation from potential linear trend. Our detrended model will provide useful for further decomposition or static testing. We will detrend our data by subtracting the fitted linear trend from observed values.

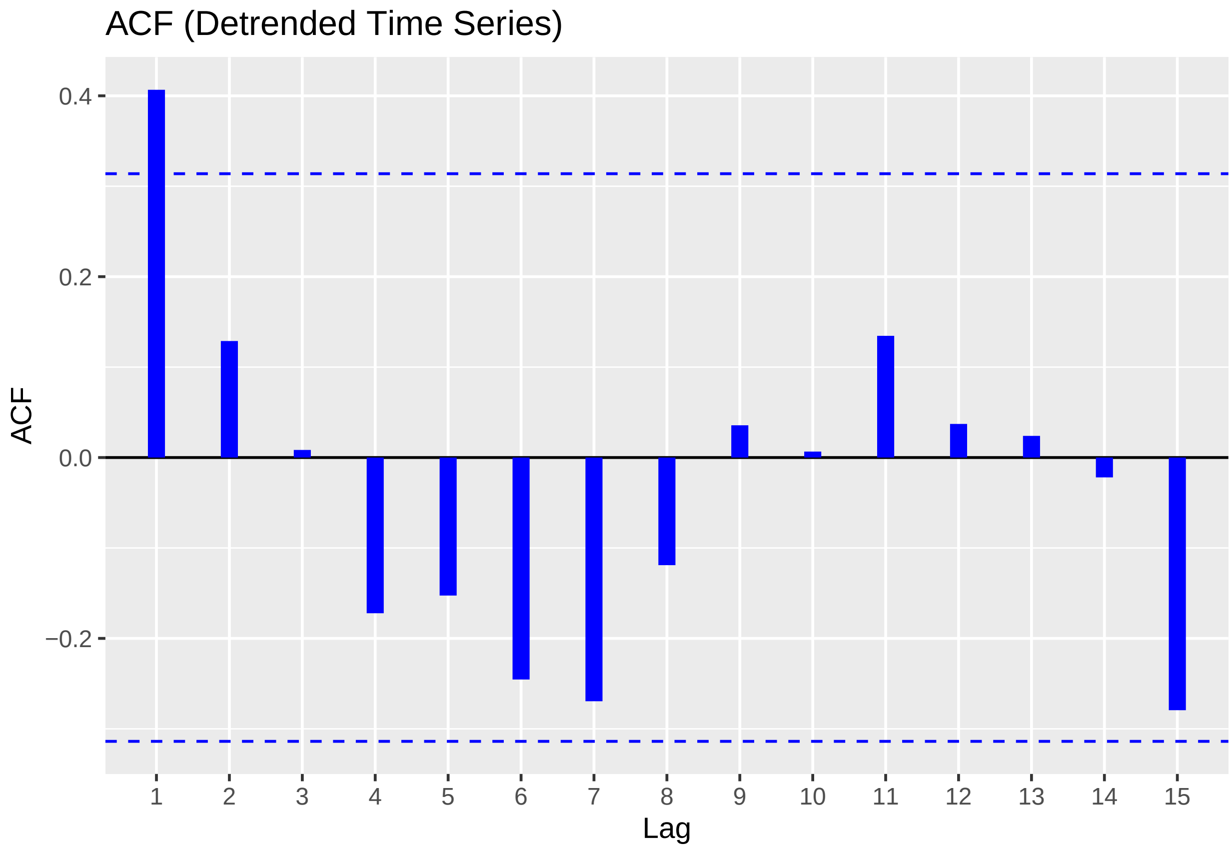
```
# detrended time series
detrended_ts <- residuals(lm_annual)
plot(
  yrs,
  detrended_ts,
  type= "o",
  main= "Detrended Yearly Time Series",
  ylab= "Detrended Value",
  xlab= "Time",
  panel.first= grid()
)
```



1.4. Make acf and pacf plots for the detrended time series to identify possible ARMA models. That is, determine the possible orders p for the AR component and the orders q for the MA component.

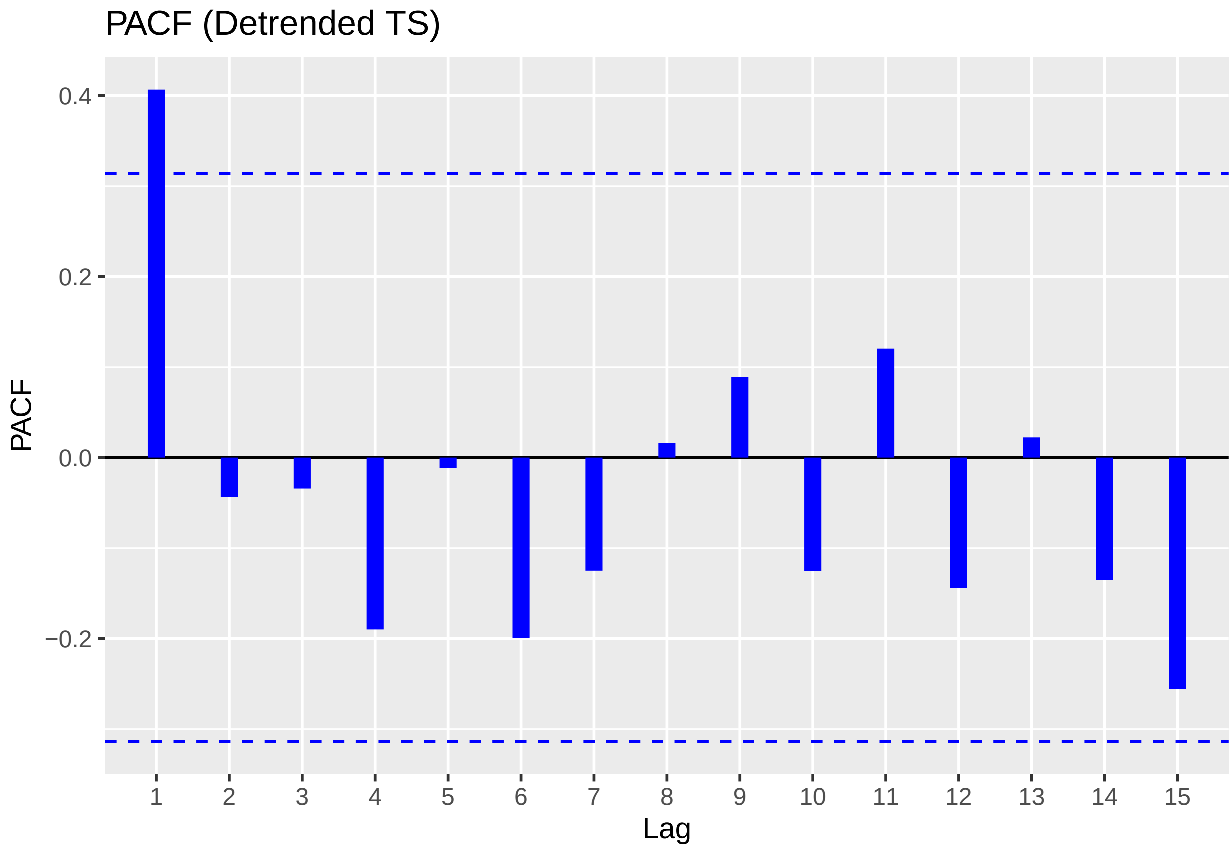
```
# ACF plot
ggAcf(detrended_ts,
      main= "ACF (Detrended Time Series)",
      size= 3, col= "blue")
```

```
## Warning in ggplot2::geom_segment(lineend = "butt", ...): Ignoring unknown
## parameters: ‘main‘
```




```r
# PACF plot
ggPacf(detrended_ts,
       main= "PACF (Detrended TS)",
       size= 3, col= "blue")
```

```
## Warning in ggplot2::geom_segment(lineend = "butt", ...): Ignoring unknown
## parameters: ‘main‘
```



In determining a suitable ARMA model for our detrended time series, we examined the Auto-correlation and Partial auto-correlation function plot generated using residuals of the linear trend model.

ACF Observations:

```
- Shows significant spike at lag=1, which is followed by values which quickly taper off and fall within

- This cutoff after lag=1 is native to a Moving Average process, with order q=1 $\rightarrow$ suggesting
```

PACF Observations:

```
- Also shows significant spike at lag=1, however subsequent lags do not show substantial values.

- This quality is characteristic of Auto-regressive processes of order p=1 $\rightarrow$ suggests a pote
```

Model Implications:

Since both ACF and PACF show prominent spike at lag=1 and no significant subsequent autocorrelation, we can establish a model as follows:

> ARMA(1,1): Both MA(1) and AR(1) components are relavent.

However, we will also establish simpler models, such as *AR(1)* and *MA(1)* for comparing using model selection criteria (AIC/BIC).

1.5. Fit the possible ARMA models you identified and perform model selection to determine the 'best' model.

To identify our optimal ARMA model for our detrended time series, we will fit our three candidate models we found from ACF/PACF plot analysis.

```
- AR(1): Autoregressive model of order 1.

- MA(1): Moving average model of order 1.

- ARMA(1,1): Combined model autoregressive + moving average.
```

We will use the *Arima()* function from the **forecast** library to fit each ARMA model and determine optimal choice by comparing AIC/BIC values.

```r
# fit AR(1), MA(1), ARMA(1,1)
model_ar1 <- Arima(detrended_ts, order= c(1,0,0))
model_ma1 <- Arima(detrended_ts, order= c(0,0,1))
model_arma11 <- Arima(detrended_ts, order= c(1,0,1))

# summary(model_arma11)

# AIC vs BIC to pick best model
model_comparison <- data.frame(
  Model = c("AR(1)", "MA(1)", "ARMA(1,1)"),
  AIC = c(AIC(model_ar1), AIC(model_ma1), AIC(model_arma11)),
  BIC = c(BIC(model_ar1), BIC(model_ma1), BIC(model_arma11))
)

print(model_comparison)
```

```
##       Model      AIC      BIC
## 1     AR(1) 63.54890 68.53959
## 2     MA(1) 63.97596 68.96665
## 3 ARMA(1,1) 65.51581 72.17006
```

Results:

```
- The AR(1) model has lowest AIC/BIC values by a small margin, indicating it offers the best trade off

- MA(1) very close in performance.

- ARMA(1,1) clearly not an optimal choice.
```

Based on model selection criteria, we will select AR(1) model as our choice for modeling this time series. This implies a linear relationship with last year's anomaly alone best explains the current year's temperature anomaly.

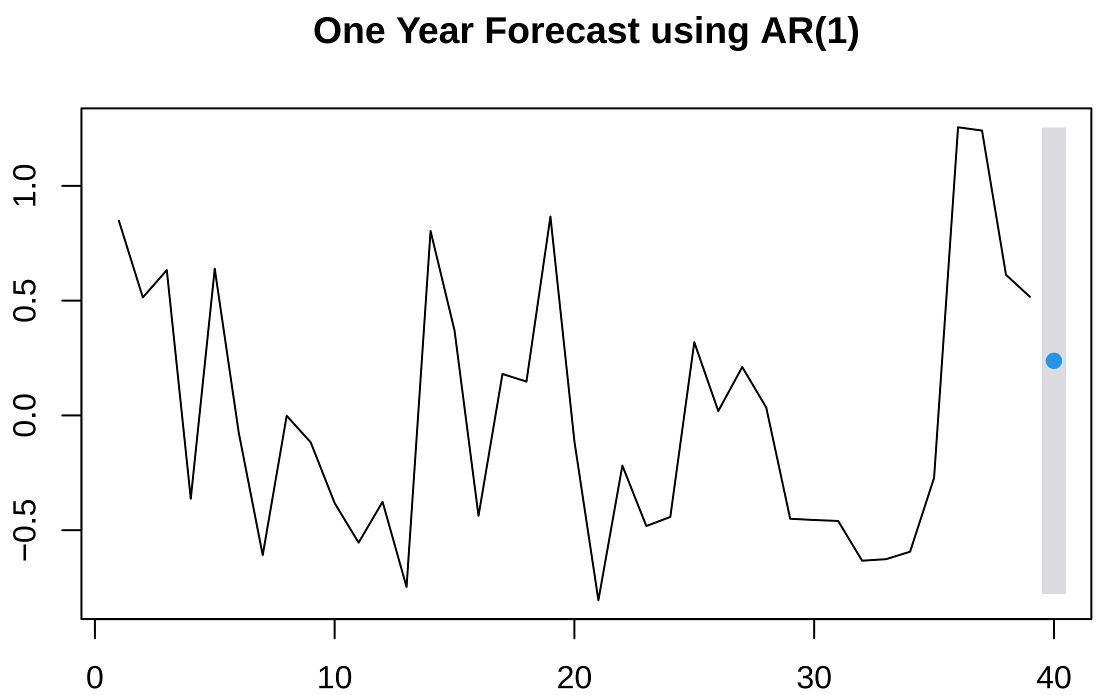
1.6. Perform a one-year-ahead forecast. Calculate the prediction and provide a 95% Prediction Interval.

Using the AR(1) model, we will perform a one year forecast to predict the next value in our detrended anomaly series. We will use the *forecast()* function with 'h=1' to compute forecast for one time step ahead, as well as specify a 95% Prediction Interval to quantify uncertainty around forecast.

```r
forecast_ar1 <- forecast(model_ar1, h= 1, level= 95)
print(forecast_ar1)
```

```
##    Point Forecast      Lo 95    Hi 95
## 40      0.2378151 -0.7777692 1.253399
```

```r
plot(forecast_ar1, main= "One Year Forecast using AR(1)")
```



Forecasting Results:

We can predict a temperature anomaly for next year to be approximately 0.2378 °C. Our 95% Prediction Interval ranges from −0.778 °C to 1.253 °C, meaning we are 95% confident values will fall in this range. The positive point forecast alludes a mild increase in temperature anomaly which continues slow upwards patter of recent years. The wide nature of the prediction interval can be reflective of external factors that our model could not account for, and also highlights uncertainty in long term forecasting from lower order AR models, which prove to be more useful for short term predictions.

Problem 2. Analyzing Monthly Average Temperature Time Series

Repeat the analysis for Problem 1 but for the monthly average time series (which canbe extracted using the script below). An important component of this problem is the analysis of seasonal patterns when dealing with sub-annual data. Some .yiytechniques from Week 14 should be employed to complete this problem.

2.2. Describe the main features of the monthly time series data you extracted.

```
time_series_monthly_matrix <- tapply(time_series_daily, list(mon, yr), mean)
time_series_monthly <- ts(as.vector(time_series_monthly_matrix), start = c(min(yr), min(mon)), frequency
monthly_features <- tsfeatures(time_series_monthly)

print("Monthly Time Series summary: ")
```

```
## [1] "Monthly Time Series summary: "
```

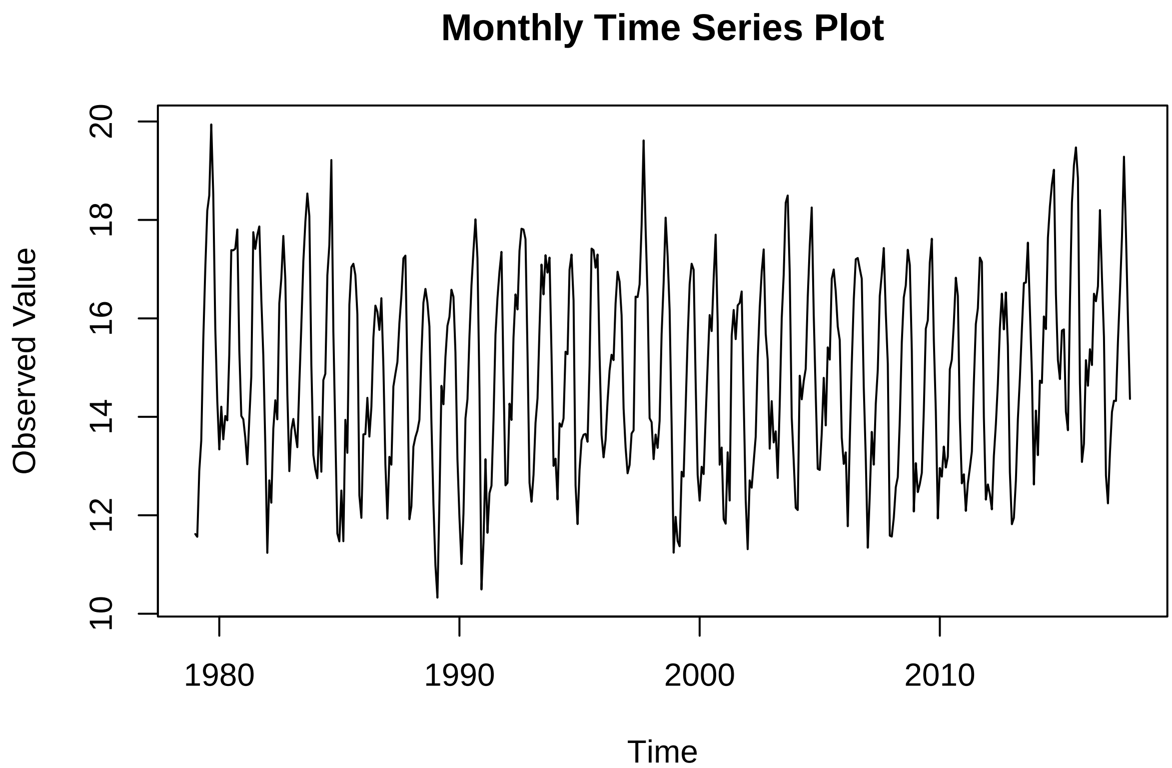
```
print(summary(time_series_monthly))
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   10.33   13.23   14.77   14.86   16.45   19.94
```

```
print(monthly_features)
```

```
## # A tibble: 1 x 20
##   frequency nperiods seasonal_period trend      spike linearity curvature e_acf1
##       <dbl>    <dbl>           <dbl> <dbl>      <dbl>     <dbl>     <dbl>  <dbl>
## 1        12        1              12 0.506    8.56e-8     0.555      2.55  0.208
## # i 12 more variables: e_acf10 <dbl>, seasonal_strength <dbl>, peak <dbl>,
## #   trough <dbl>, entropy <dbl>, x_acf1 <dbl>, x_acf10 <dbl>, diff1_acf1 <dbl>,
## #   diff1_acf10 <dbl>, diff2_acf1 <dbl>, diff2_acf10 <dbl>, seas_acf1 <dbl>
```

```
plot(time_series_monthly, main = "Monthly Time Series Plot", ylab = "Observed Value", xlab = "Time")
```



Our time series feature analysis using *tsfeatures* for the monthly series reveals the following:

Feature	Value	Analysis
frequency	12	Shows strong monthly and seasonal patterns - as expected in climate data.
season	12	Shows strong monthly and seasonal patterns - as expected in climate data.
trend	0.50617	Moderate trend can be detected.
spike	8.56e-08	Extremely low, shows no outliers in time series data.
linearity	0.555405	Points to nonlinearity, meaning other patterns such as seasonal or cyclical could be at play.
curvature	2.554952	Points to nonlinearity, meaning other patterns such as seasonal or cyclical could be at play.
entropy	0.497834	Shows high levels of randomness/unpredicability.

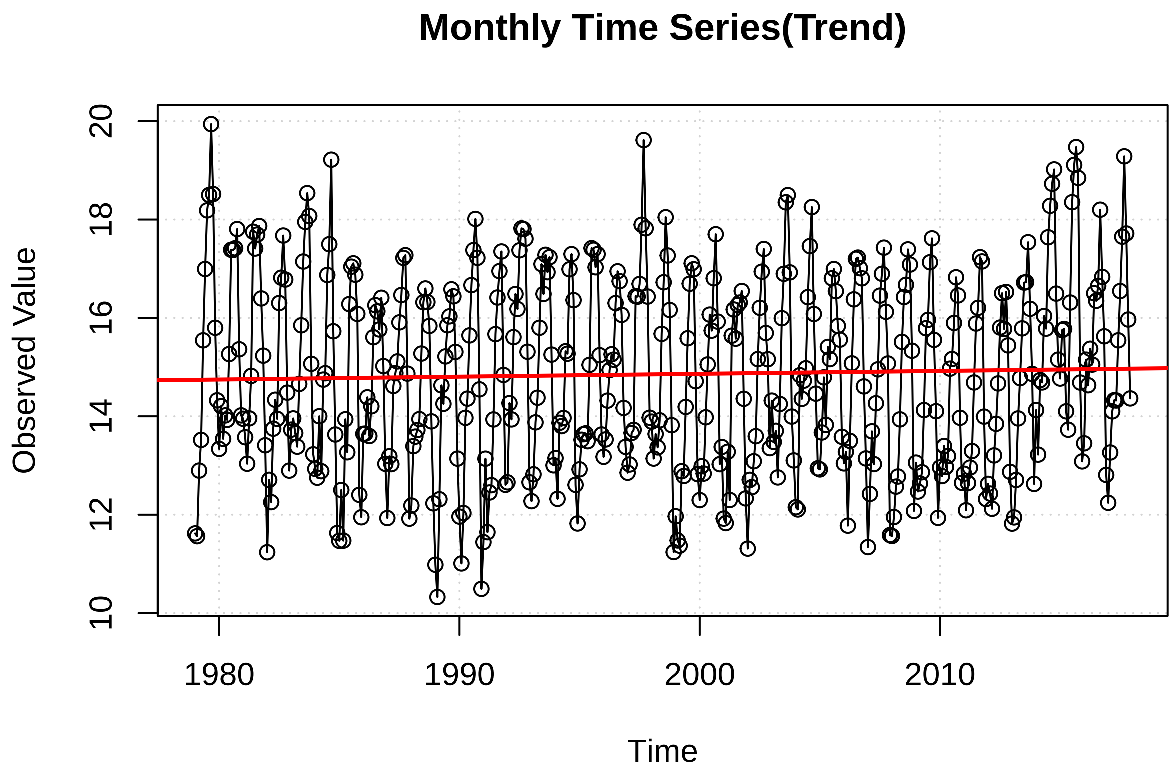
The Monthly TS Plot clearly shows patterns of seasonality - attributed to the periodic peaks and drops

2.3. Is it reasonable to assume that there is a linear trend? If so, estimate and remove the trend to get the detrended time series.

```
time_index <- time(time_series_monthly)
lm_monthly <- lm(time_series_monthly ~ time_index)

summary(lm_monthly)
```

```
##
## Call:
## lm(formula = time_series_monthly ~ time_index)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -4.4740 -1.6431 -0.1058  1.5824  5.1941
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  3.239169  16.110097   0.201    0.841
## time_index   0.005813   0.008061   0.721    0.471
##
## Residual standard error: 1.963 on 466 degrees of freedom
## Multiple R-squared:  0.001114,   Adjusted R-squared:  -0.001029
## F-statistic: 0.5199 on 1 and 466 DF,  p-value: 0.4712
```

```r
plot(
  x= time_index,
  y= time_series_monthly,
  type= "o", # measured points w/ time continuity
  main= "Monthly Time Series(Trend)",
  ylab= "Observed Value",
  xlab= "Time",
  panel.first= grid()
  )
abline(lm_monthly, col = "red", lwd = 2)
```

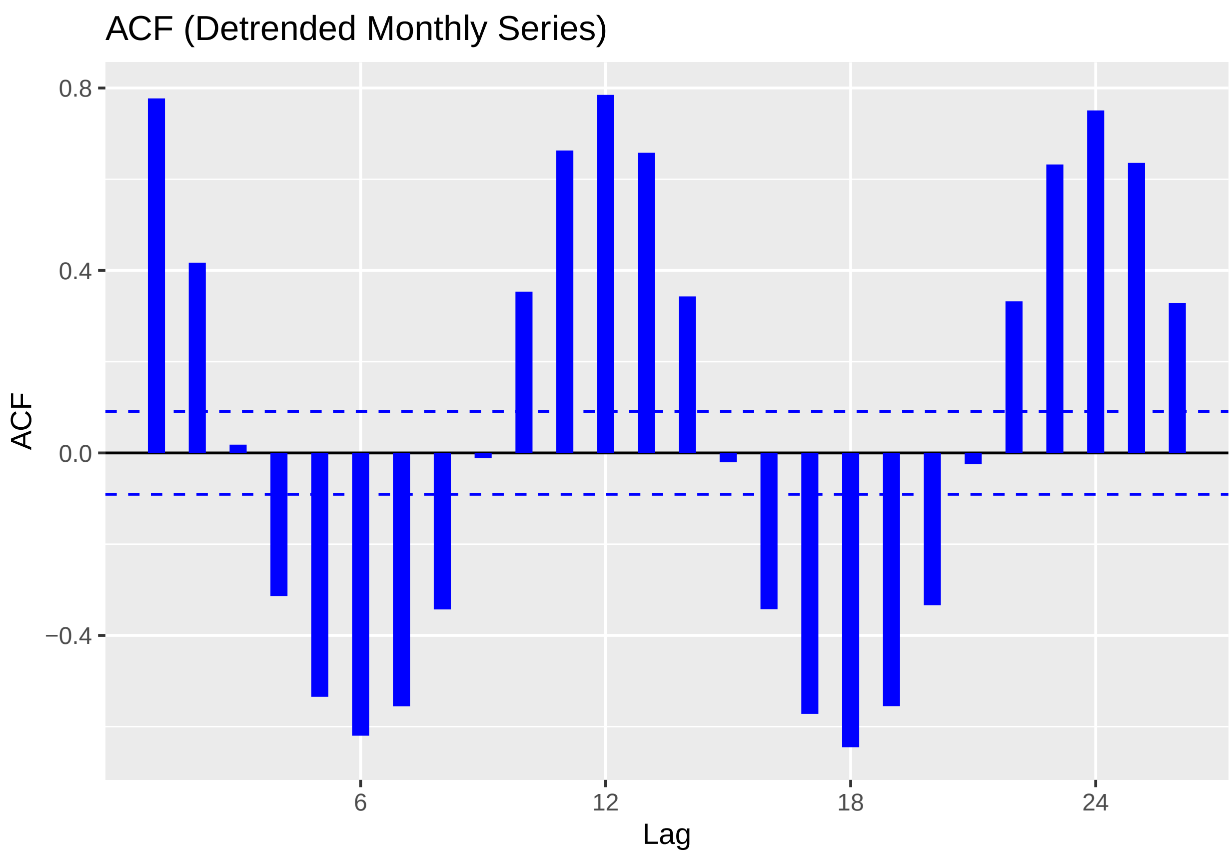


There is no strong evidence that the series shows any type of linear trend as it's p-value is very high (0.4712). This is reflected in the plot, where points rarely follow a linear pattern. Thus we will not go through the process of detrending the time series.

2.4. Make acf and pacf plots for the detrended time series to identify possible ARMA models. That is, determine the possible orders p for the AR component and the orders q for the MA component.

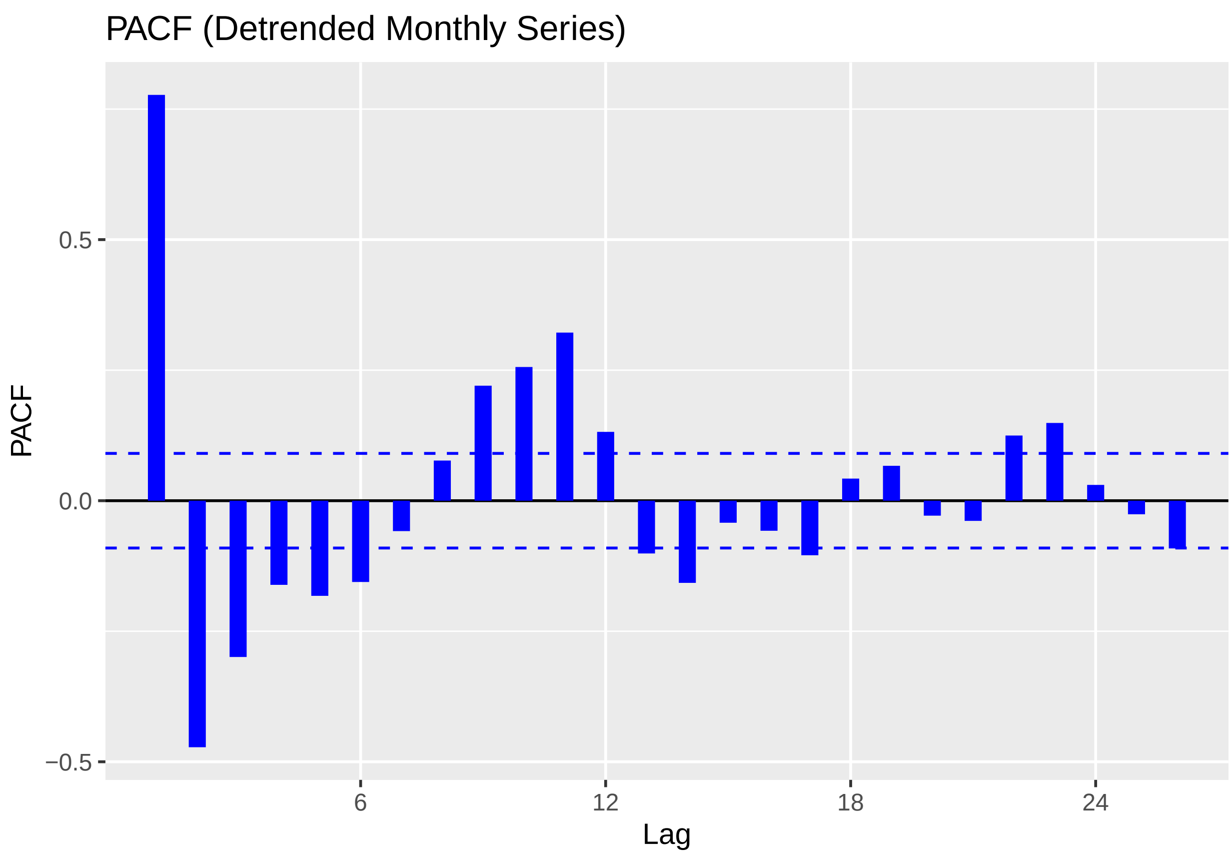
```r
# ACF plot
ggAcf(time_series_monthly, main = "ACF (Detrended Monthly Series)", size = 3, col = "blue")
```

```
## Warning in ggplot2::geom_segment(lineend = "butt", ...): Ignoring unknown
## parameters: `main`
```




```r
# PACF plot
ggPacf(time_series_monthly, main = "PACF (Detrended Monthly Series)", size = 3, col = "blue")
```

```
## Warning in ggplot2::geom_segment(lineend = "butt", ...): Ignoring unknown
## parameters: ‘main‘
```



ACF Plot:

```
- Strong spike at lag 12, indicating seasonality.
- All other lags have relatively moderate change between lags.
- Good candidates would be MA(1), MA(2)
```

PACF Plot:

```
- Strong spike at lag 1: AR(1) candidate.
```

Our plots reveal the best candidate models for fitting are:

```
- AR(1)
- MA(1), MA(2)
- ARMA(1,1), ARMA(1,2)
```

2.5. Fit the possible ARMA models you identified and perform model selection to determine the 'best' model.

```r
# fit AR(1), MA(1/2), ARMA(1,1/2)
month_model_ar1 <- Arima(time_series_monthly, order=c(1,0,0))
month_model_ma1 <- Arima(time_series_monthly, order=c(0,0,1))
month_model_ma2 <- Arima(time_series_monthly, order=c(0,0,2))
```

```r
month_model_arma11 <- Arima(time_series_monthly, order=c(1,0,1))
month_model_arma12 <- Arima(time_series_monthly, order=c(1,0,2))


summary(month_model_arma12)
```

```
## Series: time_series_monthly
## ARIMA(1,0,2) with non-zero mean
##
## Coefficients:
##          ar1     ma1     ma2     mean
##       0.5444  0.5037  0.3155  14.8388
## s.e.  0.0535  0.0571  0.0479   0.2024
##
## sigma^2 = 1.222:  log likelihood = -709.7
## AIC=1429.4   AICc=1429.53   BIC=1450.14
##
## Training set error measures:
##                      ME     RMSE       MAE        MPE     MAPE      MASE
## Training set 0.00451557 1.100839 0.8932414 -0.5235271 6.158352 0.9619373
##                    ACF1
## Training set 0.03384747
```

```r
# AIC vs BIC to pick best model
model_comparison <- data.frame(
  Model = c("AR(1)", "MA(1)", "MA(2)", "ARMA(1,1)", "ARMA(1,2)"),
  AIC = c(AIC(month_model_ar1), AIC(month_model_ma1), AIC(month_model_ma2), AIC(month_model_arma11), AIC
  BIC = c(BIC(month_model_ar1), BIC(month_model_ma1), BIC(month_model_ma2), BIC(month_model_arma11), BIC
)

print(model_comparison)
```

```
##       Model      AIC      BIC
## 1     AR(1) 1527.159 1539.604
## 2     MA(1) 1631.415 1643.860
## 3     MA(2) 1484.965 1501.559
## 4 ARMA(1,1) 1464.298 1480.892
## 5 ARMA(1,2) 1429.396 1450.138
```

AMRA(1,2) achieved the lowest AIC/BIC criterion values, outperforming our AR(1) MA(1/2) and ARMA(1,1) models. The reason for its sucsess is likely its increased flexibility useful for seasonal/cyclical data. The specific choice in ARMA(1,2) indicates that the monthly time series shows short term autocorrelation (achived by AR(1)) as well as requires corrections for noise across two periods (MA(2)) showing slightly longer dependence in the series further than short term immediate months.

The significantly higher AIC/BIC values compared to the yearly series once again confirms increased complexity in the monthly series. This is expected as monthly data naturally contains more variation/short-term rapid changes which can't be captured by simpler models like AR(1), MA(1), MA(2), or even ARMA(1,1).

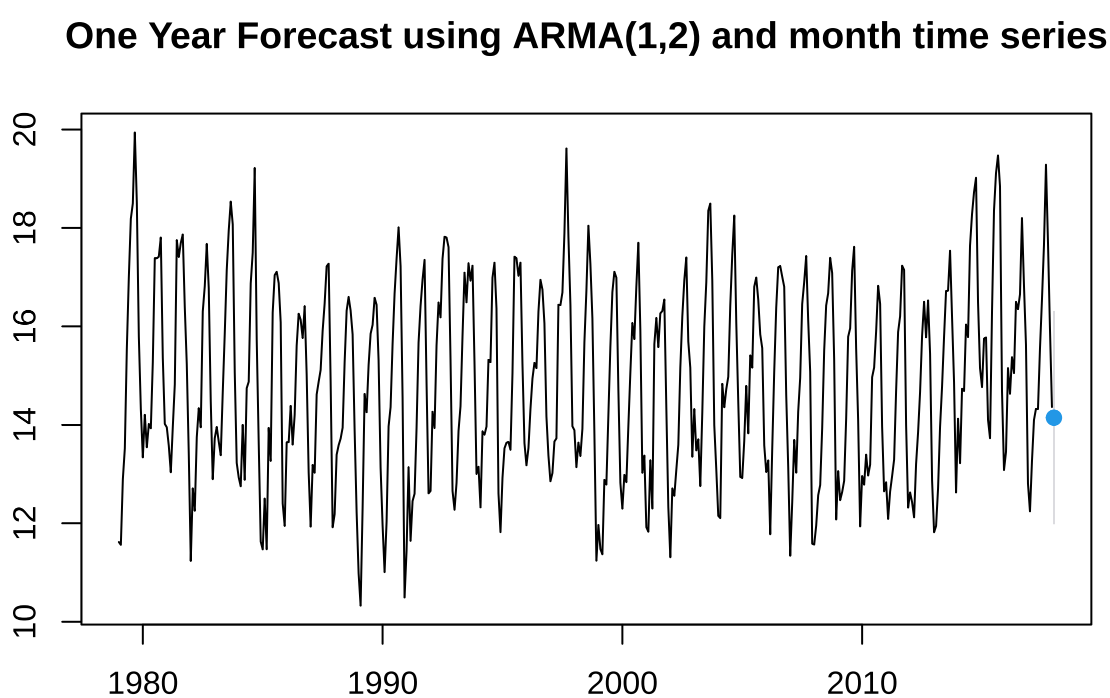
ARMA(1,2) is the selected model moving forward.

2.6. Perform a one-year-ahead forecast. Calculate the prediction and provide a 95% Prediction Interval.

```
month_forecast_arma12 <- forecast(month_model_arma12, h= 1, level= 95)
print(month_forecast_arma12)
```

```
##          Point Forecast    Lo 95    Hi 95
## Jan 2018       14.14525 11.97836 16.31213
```

```
plot(month_forecast_arma12, main= "One Year Forecast using ARMA(1,2) and month time series")
```



Problem 3. Spatial Interpolation Using Gaussian process model.

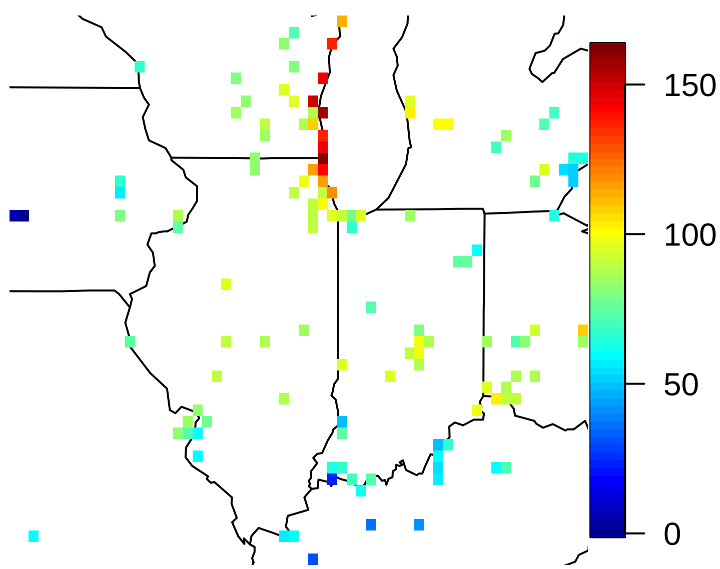
In this problem we will use the fields package to conduct spatial interpolation using Gaussian process model. We use ozone2 dataset in the fields package to interpolate average ozone levels.

3.1. Use the following script to load the data:

```
data(ozone2)
loc <- ozone2$lon.lat
rg <- apply(loc, 2, range)
y <- ozone2$y[16,]
good <- !is.na(y)
```

3.2. Visualize the spatial data using the script below and describe your findings:

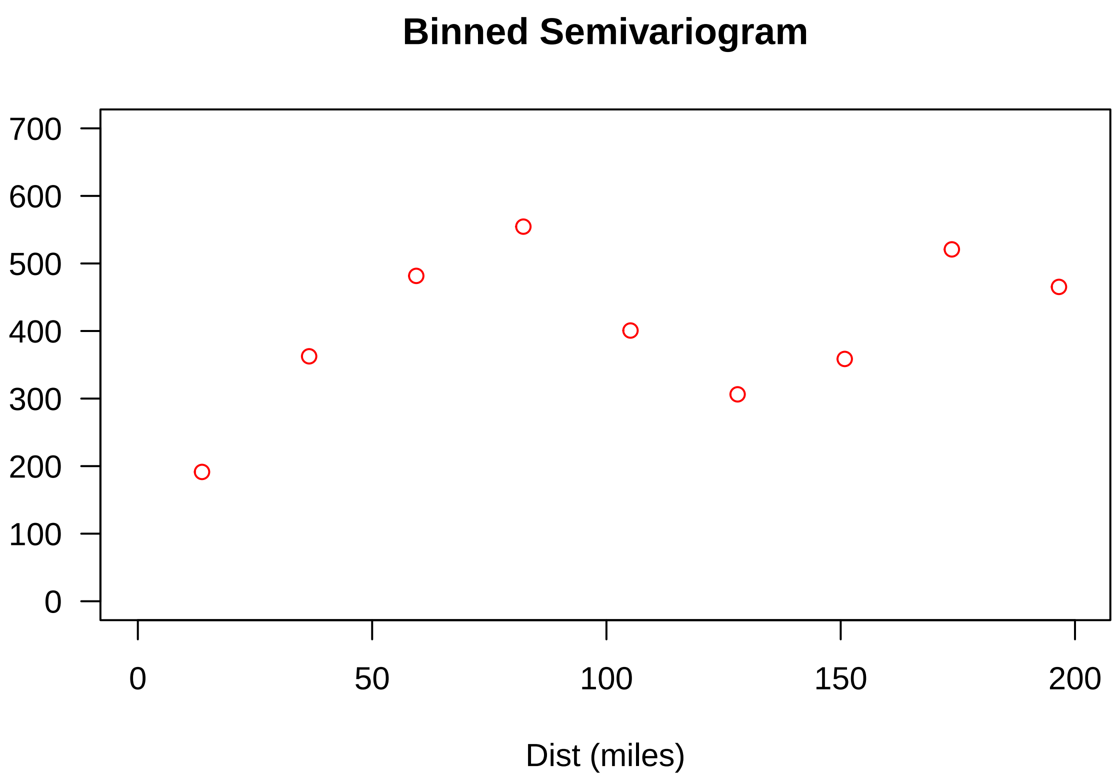
```
library(maps)
map("state", xlim = rg[, 1], ylim = rg[, 2])
quilt.plot(loc[good,], y[good], nx = 60, ny = 48, add = T)
```



3.3. Make a variogram by assuming a linear spatial trend. You may use the following script to plot the variogram. Describe qualitatively what you learned about the spatial correlation from this variogram.

```
# remove linear spatial trend
lm <- lm(y[good] ~ loc[good,])
```

```r
d <- rdist.earth(loc[good,]); maxd <- max(d)
vgram <- vgram(loc = loc[good,], y = lm$residuals, N = 30, lon.lat = TRUE)
plot(vgram$stats["mean",] ~ vgram$centers,
     main = "Binned Semivariogram",
     las = 1,
     ylab = "",
     xlab = "Dist (miles)",
     col = "red",
     xlim = c(0, 0.3 * maxd),
     ylim = c(0, 700)
     )
```



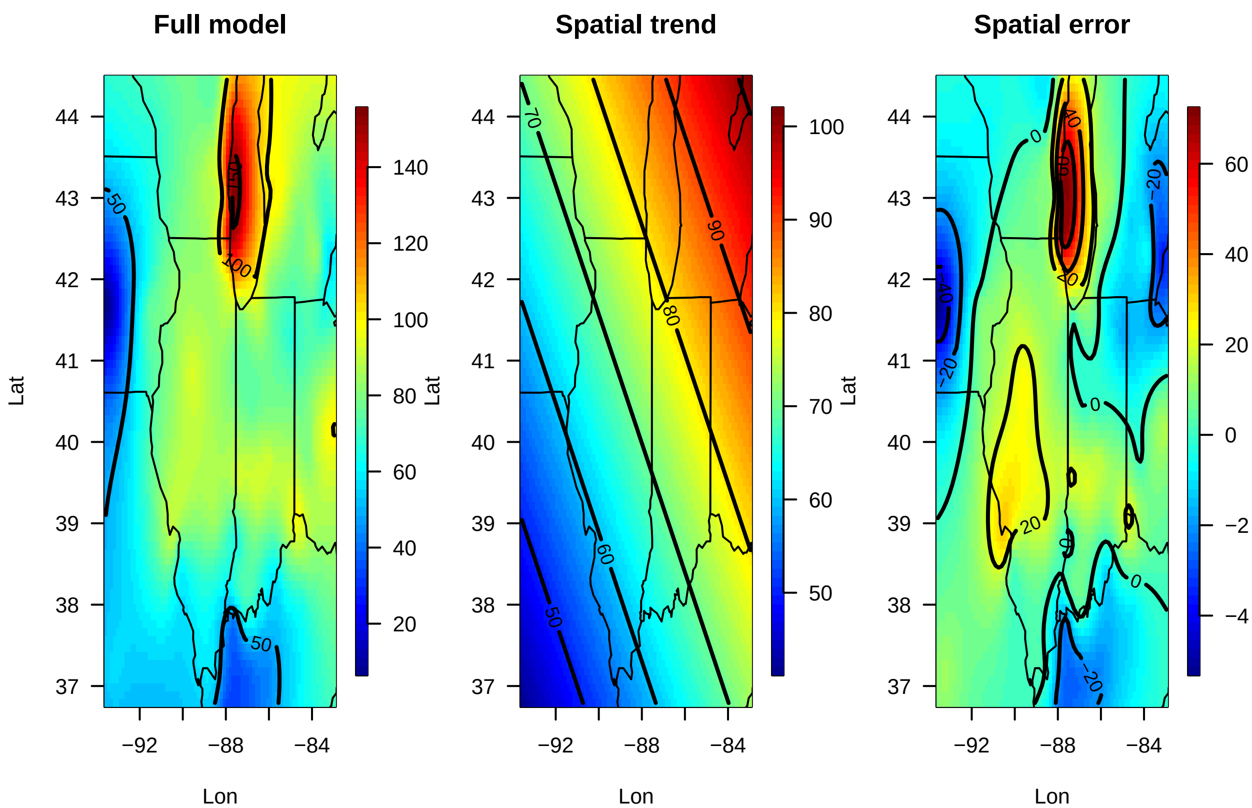
3.4. Use the following script to fit a Gaussian process to the data and to decompose the prediction into a spatial trend part and spatially correlated error part. Explain what mean function and covariance function are being used here.

```r
fit <- spatialProcess(loc[good,], y[good])

out.full <- predictSurface(fit, extrap = T)
out.poly <- predictSurface(fit, just.fixed = T, extrap = T)

out.spatial <- out.full
out.spatial$z <- out.full$z - out.poly$z
```

```r
set.panel(1, 3)
```

```
## plot window will lay out plots in a 1 by 3 matrix
```

```r
surface(out.full, las = 1, xlab = "Lon", ylab = "Lat", col = tim.colors())
title("Full model")
map("state", add = T)
surface(out.poly, las = 1, xlab = "Lon", ylab = "Lat", col = tim.colors())
map("state", add = T)
title("Spatial trend")
surface(out.spatial, las = 1, xlab = "Lon", ylab = "Lat", col = tim.colors())
map("state", add = T)
title("Spatial error")
```



3.5. Make prediction map and the associated prediction uncertainty map using the script below. Can you explain the spatial variation in the uncertainty map?

```r
xg <- fields.x.to.grid(loc[good,])
Pred <- predict.mKrig(fit, xnew = expand.grid(xg))
SE <- predictSE(fit, xnew = expand.grid(xg))

par(mfrow = c(1, 2), las = 1)
```

```r
map("state", xlim = rg[, 1], ylim = rg[, 2])
image.plot(xg[[1]], xg[[2]], matrix(Pred, 80), add = T)
map("state", xlim = rg[, 1], ylim = rg[, 2], add = T)
title("Prediction")
map("state", xlim = rg[, 1], ylim = rg[, 2])
image.plot(xg[[1]], xg[[2]], matrix(SE, 80), add = T)
map("state", xlim = rg[, 1], ylim = rg[, 2], add = T)
title("Prediction SE")
points(loc, pch = 16, cex = 0.5)
```